

Unit A4: Machine Learning (HL Extension) - Expanded Guide

Sub-topics: A4.2, A4.3

1. Syllabus Scope & Learning Outcomes

These topics are for Higher Level students only. The content moves from concepts to specific algorithms and mechanics.

Syllabus Point	Command Term	Student Expectation (The "Do")
A4.2.1	Describe	Describe Data Cleaning: Handling Outliers , Missing Data (Imputation/Deletion), Duplicates , and Normalization (Scaling).
A4.2.2	Describe	Describe Feature Selection: Filter , Wrapper , and Embedded methods to choose relevant variables.
A4.2.3	Describe	Describe Dimensionality Reduction: Reducing variables to avoid the Curse of Dimensionality (Overfitting, sparsity).
A4.3.1	Explain	Explain Linear Regression : Predicting continuous outcomes using the line of best fit ($y =$). Evaluation: r^2 .
A4.3.2	Explain	Explain Classification : K-Nearest Neighbours (K-NN) (voting based on distance) and Decision

		Trees.
A4.3.3	Explain	Explain Hyperparameter Tuning : Optimizing model settings (e.g., k in K-NN). Evaluation: Accuracy, Precision, Recall, F1 Score.
A4.3.4	Describe	Describe Clustering : Grouping data by feature similarity (Unsupervised). Example: Customer Segmentation.
A4.3.5	Describe	Describe Association Rules : Discovering relations ("If X, then Y"). Example: Market Basket Analysis.
A4.3.6	Describe	Describe Reinforcement Learning : Agent, Environment, State, Action, Reward, Policy. Exploration vs. Exploitation.
A4.3.7	Describe	Describe Genetic Algorithms : Selection, Crossover, Mutation, Fitness Function (Natural Selection analogy).
A4.3.8	Outline	Outline ANNs (Artificial Neural Networks): Input Layer, Hidden Layers, Weights, Bias, Activation Function, Output.
A4.3.9	Describe	Describe CNNs (Convolutional Neural Networks): Convolution Layers, Pooling Layers, Kernels/Filters. Used for

		Image Recognition.
A4.3.10	Explain	Explain Model Selection : Choosing the right algorithm based on data size, complexity, and desired outcome (Accuracy vs. Speed).

2. Key Terminology & Definitions

- **Preprocessing**: Transforming raw data into a clean format suitable for ML.
- **Imputation**: Replacing missing data with a calculated value (e.g., Mean/Median).
- **Normalization**: Scaling data to a specific range (e.g., 0 to 1) so large values don't dominate.
- **Overfitting**: When a model learns the training data *too* well (including noise) and fails on new data.
- **Underfitting**: When a model is too simple to capture the pattern.
- **Hyperparameters**: Settings chosen *before* training (e.g., Number of trees, Learning rate) vs. Parameters learned *during* training (Weights).
- **Confusion Matrix**: A table used to evaluate classification models (True Positives, False Positives, etc.).
- **Perceptron**: The simplest unit of a neural network (Input $\rightarrow$ Weight + Bias $\rightarrow$ Activation $\rightarrow$ Output).
- **Backpropagation**: The method used to train neural networks by calculating error and adjusting weights backwards.

3. Core Content & Exam Actions

A4.2: Data Preprocessing

Theory:

- **Cleaning**: Garbage In = Garbage Out. Must remove noise/duplicates.
- **Feature Selection**: Selecting the most useful inputs. Reduces training time and overfitting.
- **Dimensionality Reduction**: Techniques (like PCA - though PCA maths is excluded) to simplify data while keeping essential information.

Student Exam Action:

- **Scenario**: A dataset has missing ages for 5% of users. **Describe** two ways to handle this. (1. Delete those rows - easy but loses data. 2. Impute with the average age - keeps data

but adds bias).

A4.3: ML Approaches & Algorithms

Theory:

- **Linear Regression:** Used for *numbers* (e.g., Temperature). Finds the line minimizing error.
- **K-NN:** Used for *categories*. "Tell me who your neighbors are, and I'll tell you who you are."
- **Neural Networks (ANNs):** Modeled on the brain. Good for non-linear complex patterns.
- **CNNs:** specialized for grids (Images). **Convolution** extracts features (edges/textures); **Pooling** reduces size.
- **Genetic Algorithms:** Evolution simulation. create a population → test fitness → breed best (crossover) + mutate → repeat. Used for optimization (e.g., Route planning).

Student Exam Action:

- **Explanation:** Explain why a **CNN** is better for image recognition than a standard ANN. (CNNs preserve spatial relationships using filters; standard ANNs flatten images, losing the 2D structure).
- **Evaluation:** A medical model has 99% Accuracy but 0% Recall on rare diseases. Explain why **Accuracy** is a bad metric here. (It predicts "Healthy" for everyone. **Recall** is needed to ensure we catch the few sick cases).

4. Common Pitfalls & Examiner Tips

- **Pitfall (K-NN):** Forgetting to mention that k is a hyperparameter. A low k captures noise (overfitting); a high k blurs boundaries (underfitting).
- **Pitfall (CNNs):** Confusing **Convolution** (feature extraction) with **Pooling** (down-sampling/shrinking).
- **Tip (Evaluation):** Memorize: **Precision** = Quality (how many selected are relevant), **Recall** = Quantity (how many relevant are selected).